Midterm CORE part

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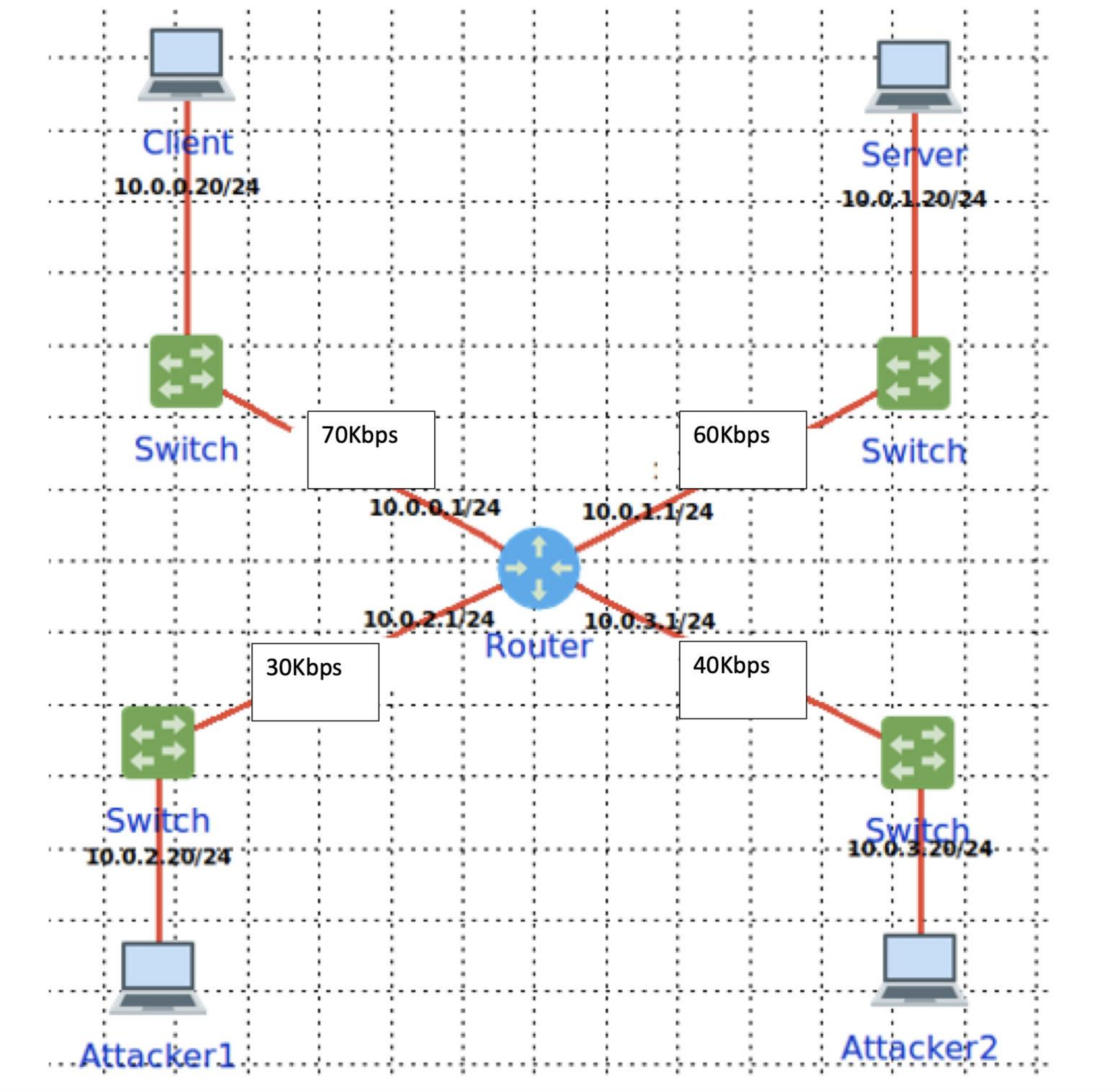
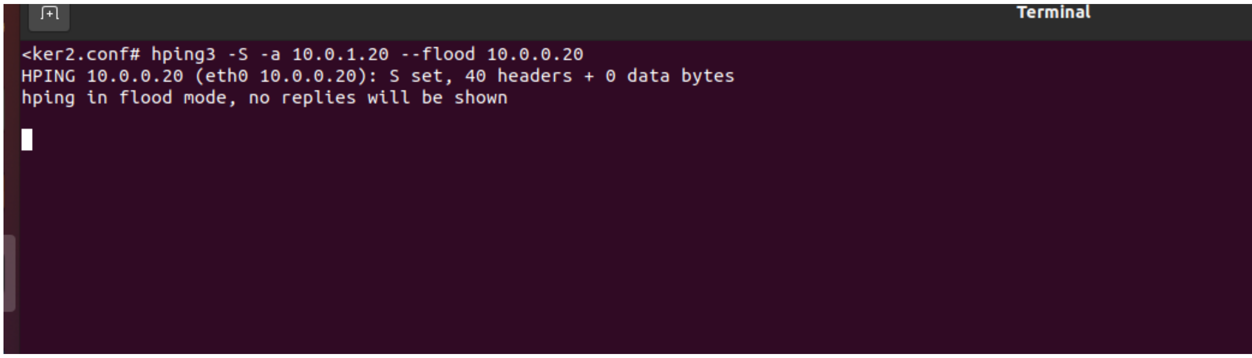
hping3 cheat sheet: https://linux.die.net/man/8/hping3,
https://cyberwar.nl/d/cheatsheets/hping3_cheatsheet_v1.0-ENG.pdf,
and https://www.kali.org/tools/hping3/

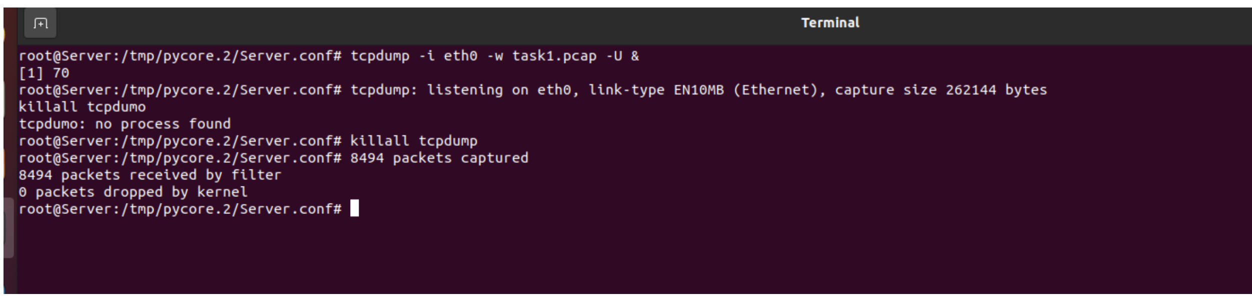
A. Using the scenario below:



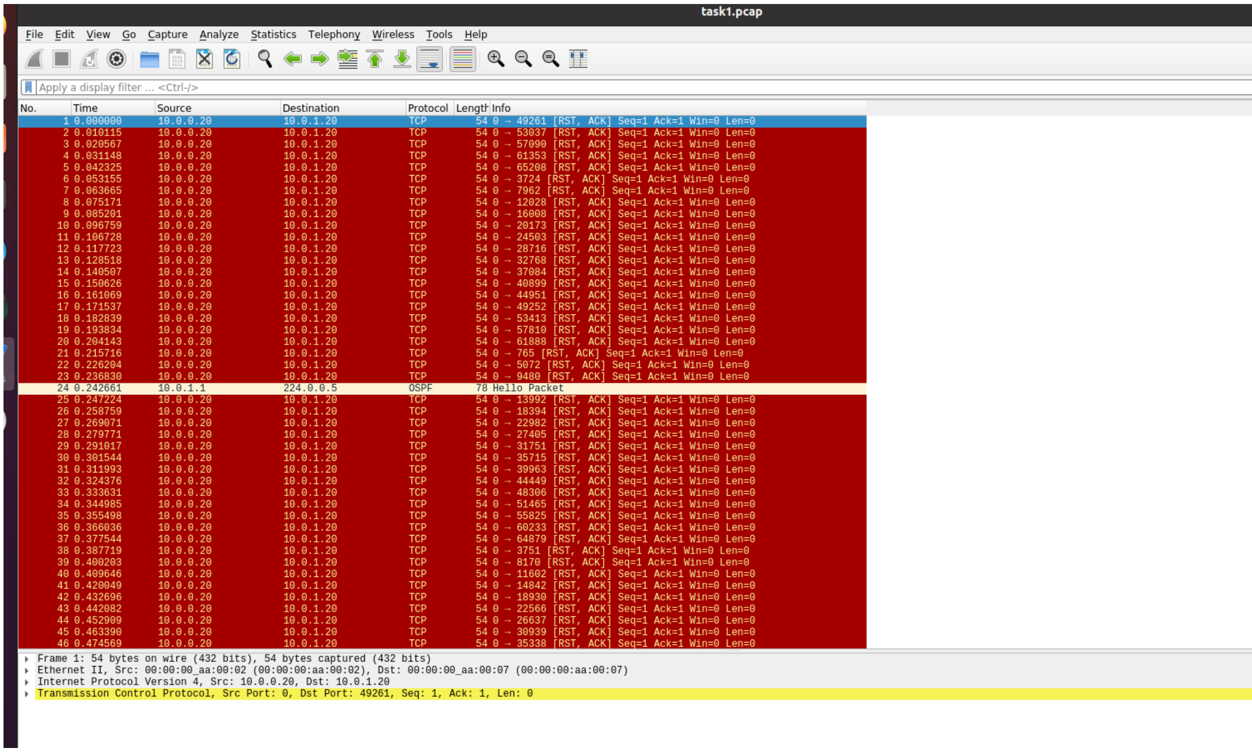
Make sure you have the exact same addresses as shown. Connect the links with the lower IP addresses first to get the same addresses

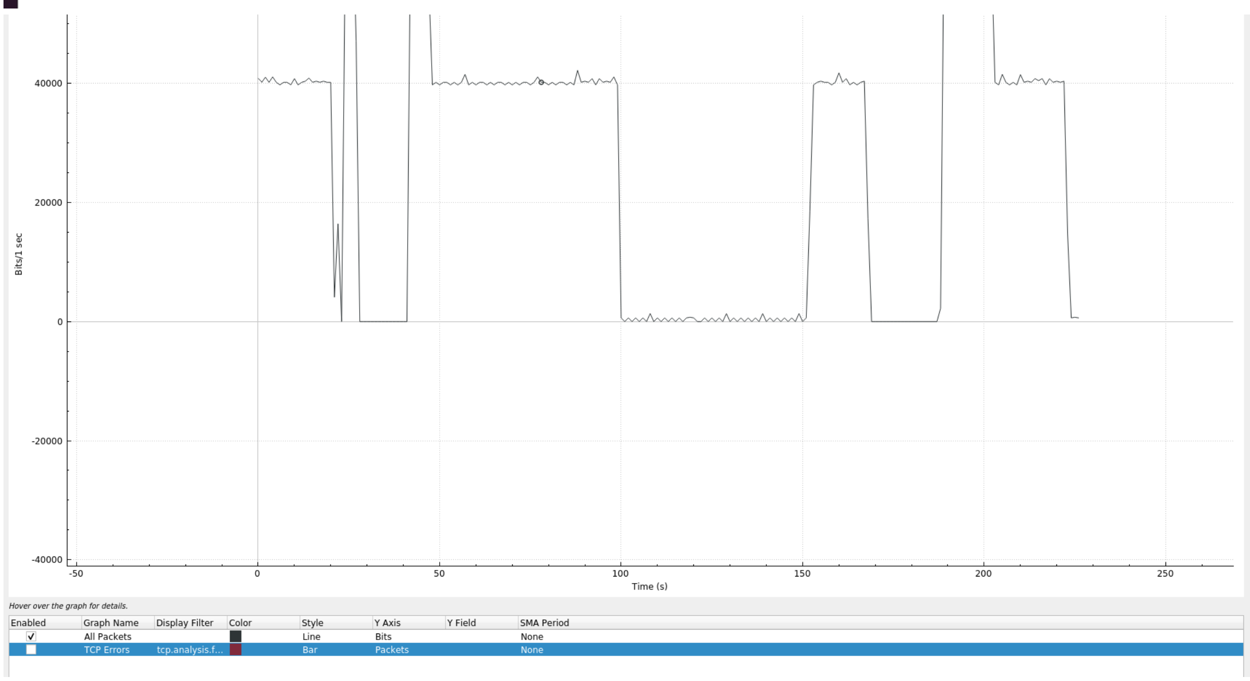
1. (1pts) Attacker 2 runs a DoS SYN reflection attack on the Server using Client as a reflection
    a. Show the exact command you used





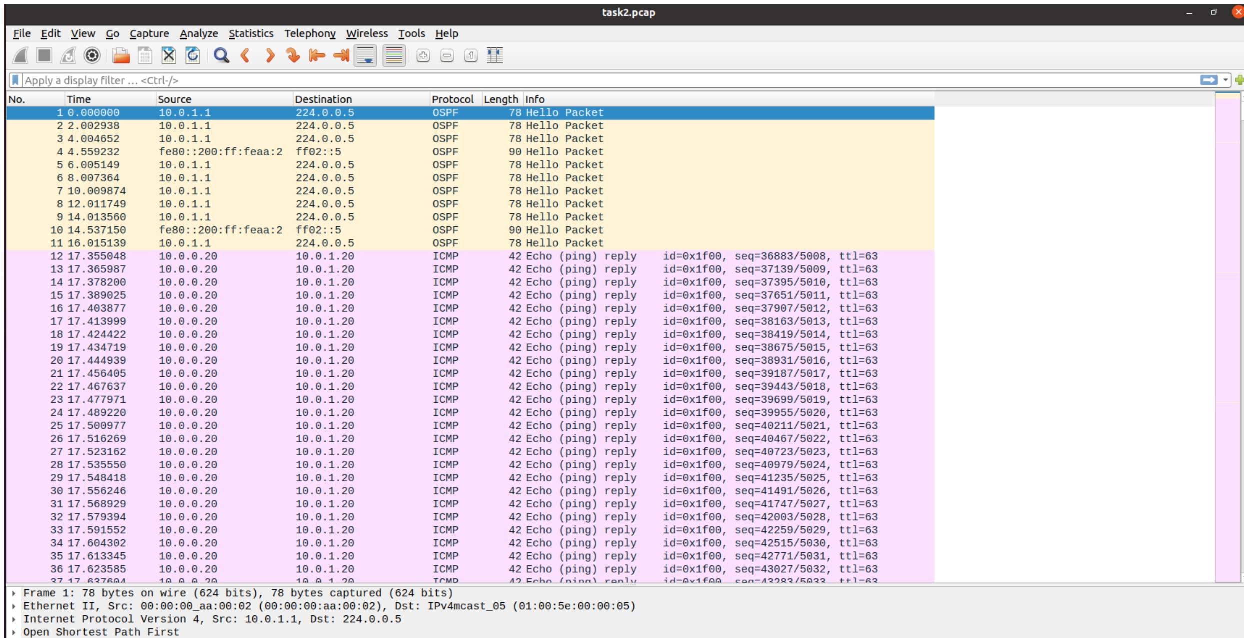
    b. What is the maximum attack magnitude?
    c. Show a Wireshark I/O graph from the Server matching your answer. **Use bits/s on the y-axis.**





   d. Describe your setup and the resulting I/O graph

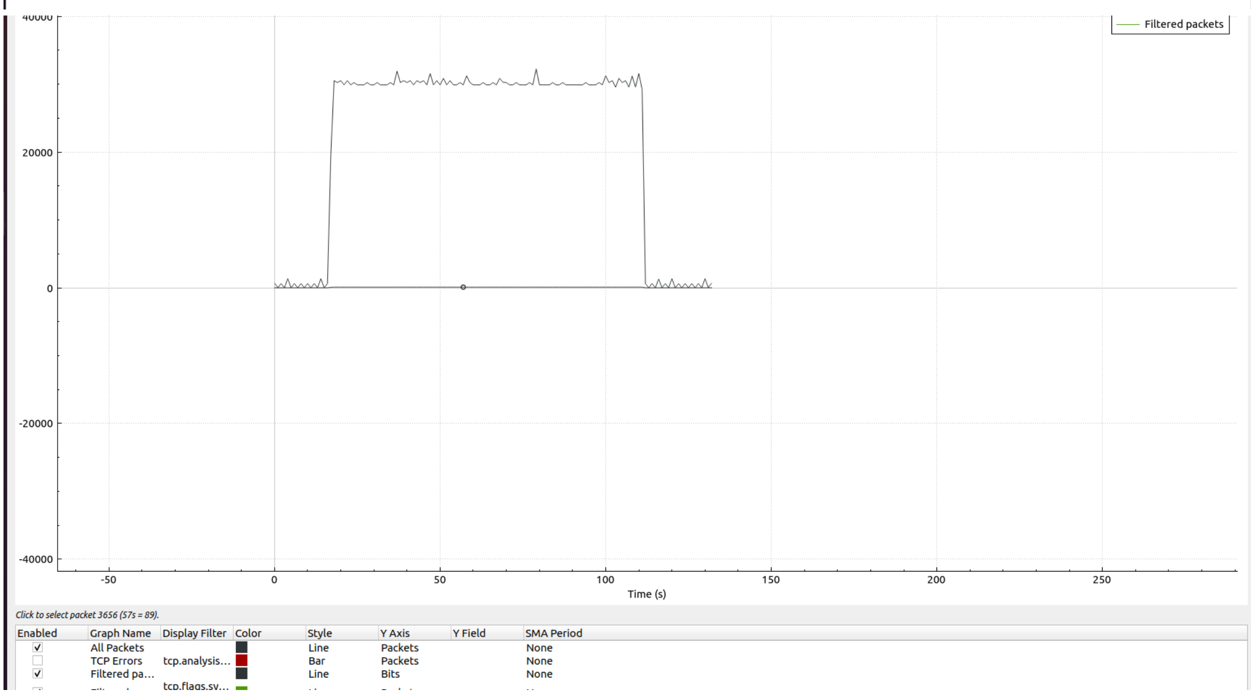
2. (1pts) Attacker 1 runs a DoS ICMP reflection attack on the Server using Client as a reflection
   a. Show the exact command you used

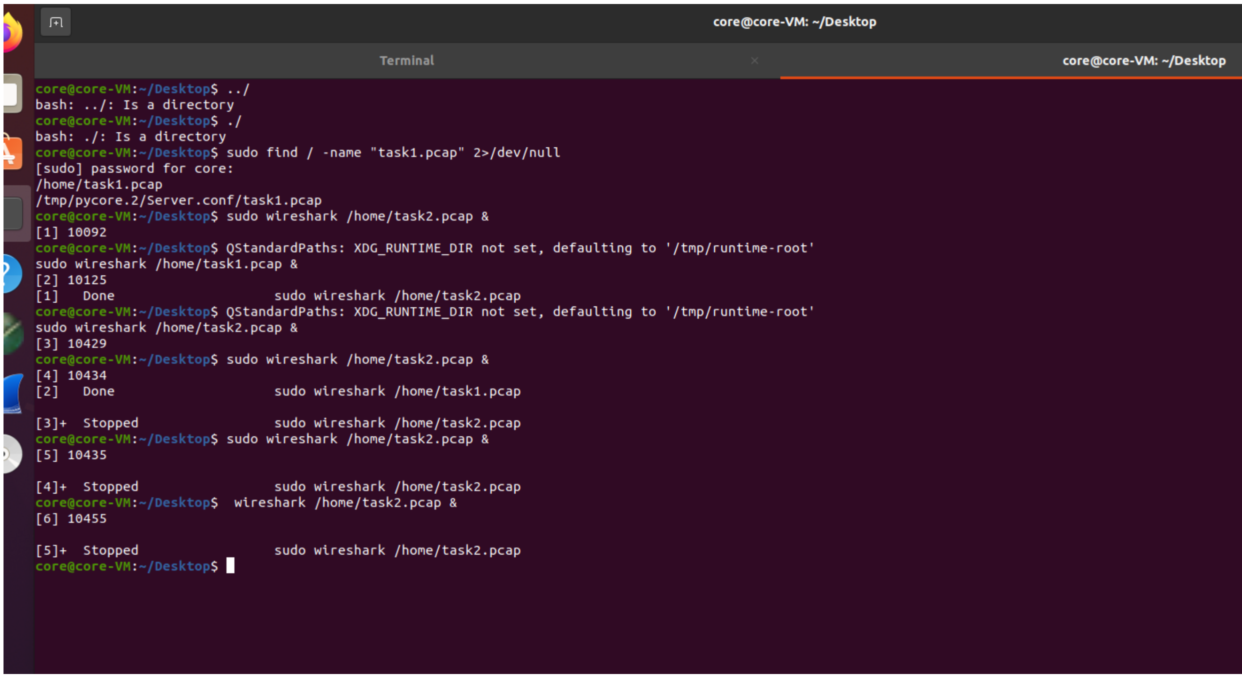


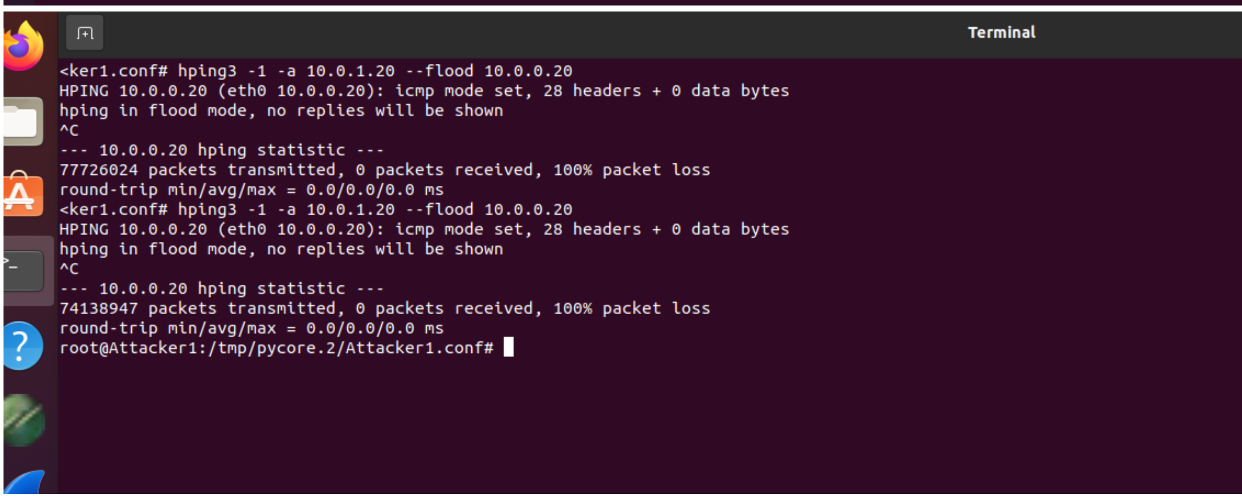
task2.pcap

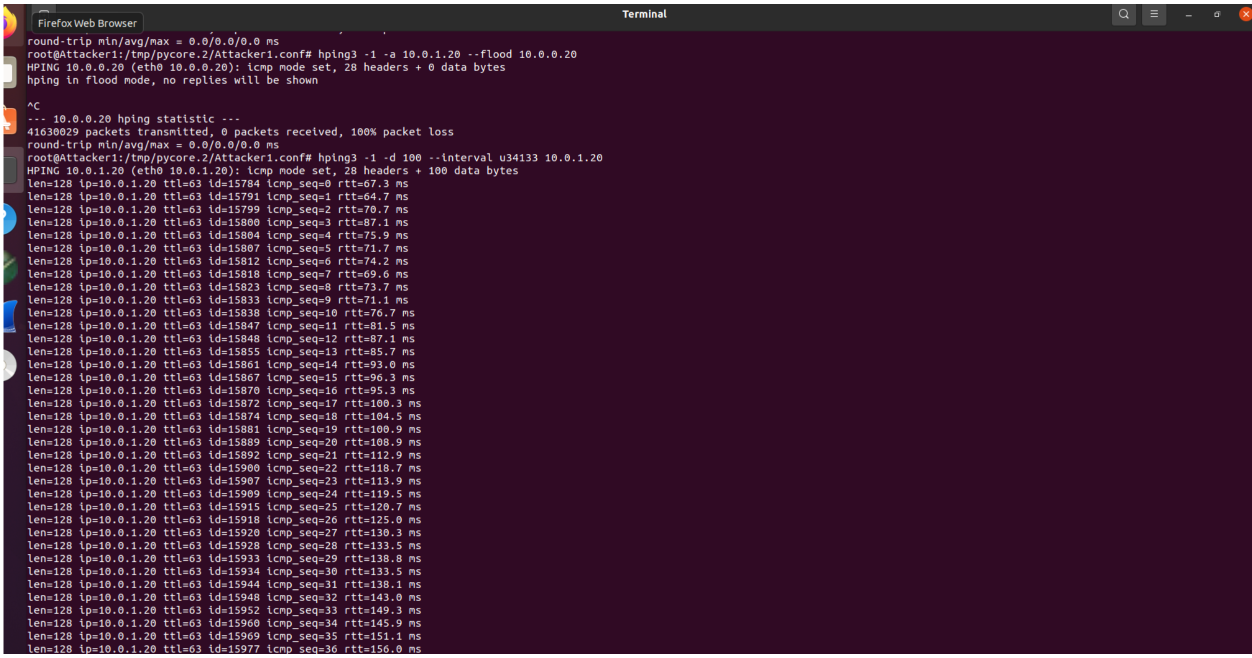
No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	10.0.1.1	224.0.0.5	OSPF	78	Hello Packet
2	2.002938	10.0.1.1	224.0.0.5	OSPF	78	Hello Packet
3	4.004652	10.0.1.1	224.0.0.5	OSPF	78	Hello Packet
4	4.559232	fe80::200:ff:feaa:2	ff02::5	OSPF	90	Hello Packet
5	6.005149	10.0.1.1	224.0.0.5	OSPF	78	Hello Packet
6	8.007364	10.0.1.1	224.0.0.5	OSPF	78	Hello Packet
7	10.009874	10.0.1.1	224.0.0.5	OSPF	78	Hello Packet
8	12.011749	10.0.1.1	224.0.0.5	OSPF	78	Hello Packet
9	14.013560	10.0.1.1	224.0.0.5	OSPF	78	Hello Packet
10	14.537150	fe80::200:ff:feaa:2	ff02::5	OSPF	90	Hello Packet
11	16.015139	10.0.1.1	224.0.0.5	OSPF	78	Hello Packet
12	17.355048	10.0.0.20	10.0.1.20	ICMP	42	Echo (ping) reply id=0x1f00, seq=36883/5008, ttl=63
13	17.365987	10.0.0.20	10.0.1.20	ICMP	42	Echo (ping) reply id=0x1f00, seq=37139/5009, ttl=63
14	17.378200	10.0.0.20	10.0.1.20	ICMP	42	Echo (ping) reply id=0x1f00, seq=37395/5010, ttl=63
15	17.389025	10.0.0.20	10.0.1.20	ICMP	42	Echo (ping) reply id=0x1f00, seq=37651/5011, ttl=63
16	17.403877	10.0.0.20	10.0.1.20	ICMP	42	Echo (ping) reply id=0x1f00, seq=37907/5012, ttl=63
17	17.413999	10.0.0.20	10.0.1.20	ICMP	42	Echo (ping) reply id=0x1f00, seq=38163/5013, ttl=63
18	17.424422	10.0.0.20	10.0.1.20	ICMP	42	Echo (ping) reply id=0x1f00, seq=38419/5014, ttl=63
19	17.434719	10.0.0.20	10.0.1.20	ICMP	42	Echo (ping) reply id=0x1f00, seq=38675/5015, ttl=63
20	17.444939	10.0.0.20	10.0.1.20	ICMP	42	Echo (ping) reply id=0x1f00, seq=38931/5016, ttl=63
21	17.456405	10.0.0.20	10.0.1.20	ICMP	42	Echo (ping) reply id=0x1f00, seq=39187/5017, ttl=63
22	17.467637	10.0.0.20	10.0.1.20	ICMP	42	Echo (ping) reply id=0x1f00, seq=39443/5018, ttl=63
23	17.477971	10.0.0.20	10.0.1.20	ICMP	42	Echo (ping) reply id=0x1f00, seq=39699/5019, ttl=63
24	17.489220	10.0.0.20	10.0.1.20	ICMP	42	Echo (ping) reply id=0x1f00, seq=39955/5020, ttl=63
25	17.500977	10.0.0.20	10.0.1.20	ICMP	42	Echo (ping) reply id=0x1f00, seq=40211/5021, ttl=63
26	17.516269	10.0.0.20	10.0.1.20	ICMP	42	Echo (ping) reply id=0x1f00, seq=40467/5022, ttl=63
27	17.523162	10.0.0.20	10.0.1.20	ICMP	42	Echo (ping) reply id=0x1f00, seq=40723/5023, ttl=63
28	17.535550	10.0.0.20	10.0.1.20	ICMP	42	Echo (ping) reply id=0x1f00, seq=40979/5024, ttl=63
29	17.548418	10.0.0.20	10.0.1.20	ICMP	42	Echo (ping) reply id=0x1f00, seq=41235/5025, ttl=63
30	17.556246	10.0.0.20	10.0.1.20	ICMP	42	Echo (ping) reply id=0x1f00, seq=41491/5026, ttl=63
31	17.568929	10.0.0.20	10.0.1.20	ICMP	42	Echo (ping) reply id=0x1f00, seq=41747/5027, ttl=63
32	17.579394	10.0.0.20	10.0.1.20	ICMP	42	Echo (ping) reply id=0x1f00, seq=42003/5028, ttl=63
33	17.591552	10.0.0.20	10.0.1.20	ICMP	42	Echo (ping) reply id=0x1f00, seq=42259/5029, ttl=63
34	17.604302	10.0.0.20	10.0.1.20	ICMP	42	Echo (ping) reply id=0x1f00, seq=42515/5030, ttl=63
35	17.613345	10.0.0.20	10.0.1.20	ICMP	42	Echo (ping) reply id=0x1f00, seq=42771/5031, ttl=63
36	17.623585	10.0.0.20	10.0.1.20	ICMP	42	Echo (ping) reply id=0x1f00, seq=43027/5032, ttl=63

Frame 1: 78 bytes on wire (624 bits), 78 bytes captured (624 bits)
Ethernet II, Src: 00:00:00_aa:00:02 (00:00:00:aa:00:02), Dst: IPv4mcast_05 (01:00:5e:00:00:05)
Internet Protocol Version 4, Src: 10.0.1.1, Dst: 224.0.0.5
Open Shortest Path First



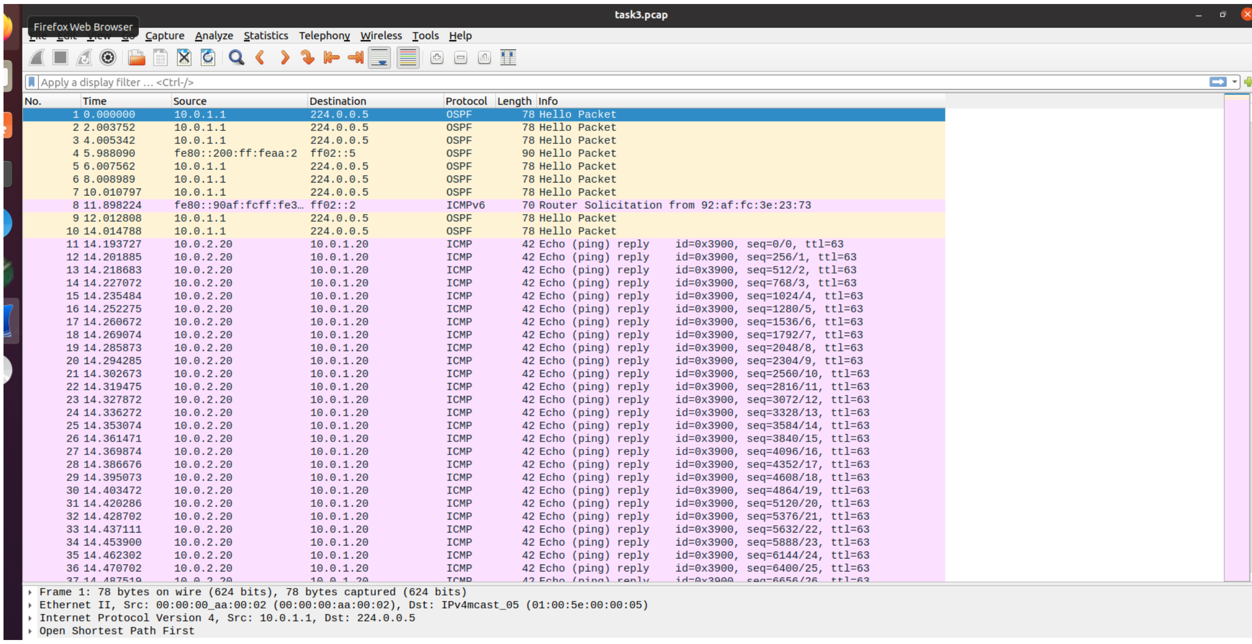

core@core-VM: ~/Desktop
Terminal
core@core-VM: ~/Desktop
core@core-VM:~/Desktop$ ../
bash: ../: Is a directory
core@core-VM:~/Desktop$ ./
bash: ./: Is a directory
core@core-VM:~/Desktop$ sudo find / -name "task1.pcap" 2>/dev/null
[sudo] password for core:
/home/task1.pcap
/tmp/pycore.2/Server.conf/task1.pcap
core@core-VM:~/Desktop$ sudo wireshark /home/task2.pcap &
[1] 10092
core@core-VM:~/Desktop$ QStandardPaths: XDG_RUNTIME_DIR not set, defaulting to '/tmp/runtime-root'
sudo wireshark /home/task1.pcap &
[2] 10125
[1]   Done   sudo wireshark /home/task2.pcap
core@core-VM:~/Desktop$ QStandardPaths: XDG_RUNTIME_DIR not set, defaulting to '/tmp/runtime-root'
sudo wireshark /home/task2.pcap &
[3] 10429
core@core-VM:~/Desktop$ sudo wireshark /home/task2.pcap &
[4] 10434
[2]   Done   sudo wireshark /home/task1.pcap
[3]+ Stopped   sudo wireshark /home/task2.pcap
core@core-VM:~/Desktop$ sudo wireshark /home/task2.pcap &
[5] 10435
[4]+ Stopped   sudo wireshark /home/task2.pcap
core@core-VM:~/Desktop$ wireshark /home/task2.pcap &
[6] 10455
[5]+ Stopped   sudo wireshark /home/task2.pcap
core@core-VM:~/Desktop$

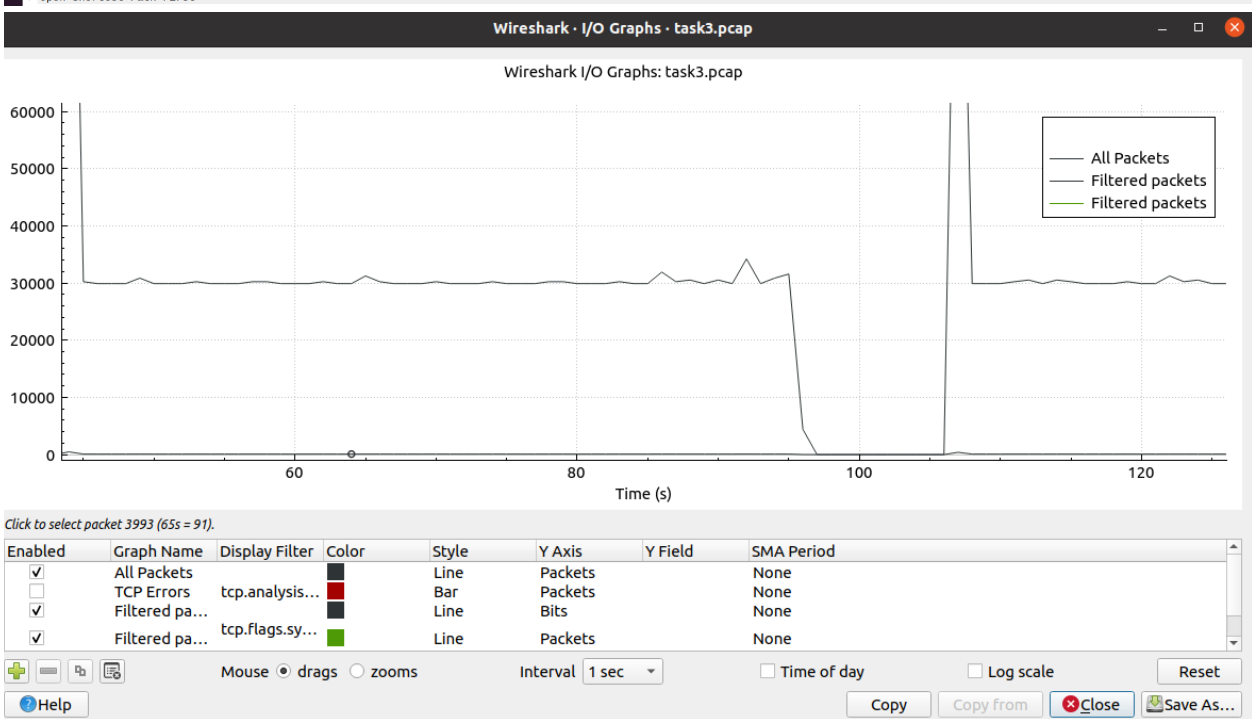

Terminal
<ker1.conf# hping3 -1 -a 10.0.1.20 --flood 10.0.0.20
HPING 10.0.0.20 (eth0 10.0.0.20): icmp mode set, 28 headers + 0 data bytes
hping in flood mode, no replies will be shown
^C
--- 10.0.0.20 hping statistic ---
77726024 packets transmitted, 0 packets received, 100% packet loss
round-trip min/avg/max = 0.0/0.0/0.0 ms
<ker1.conf# hping3 -1 -a 10.0.1.20 --flood 10.0.0.20
HPING 10.0.0.20 (eth0 10.0.0.20): icmp mode set, 28 headers + 0 data bytes
hping in flood mode, no replies will be shown
^C
--- 10.0.0.20 hping statistic ---
74138947 packets transmitted, 0 packets received, 100% packet loss
round-trip min/avg/max = 0.0/0.0/0.0 ms
root@Attacker1:/tmp/pycore.2/Attacker1.conf#

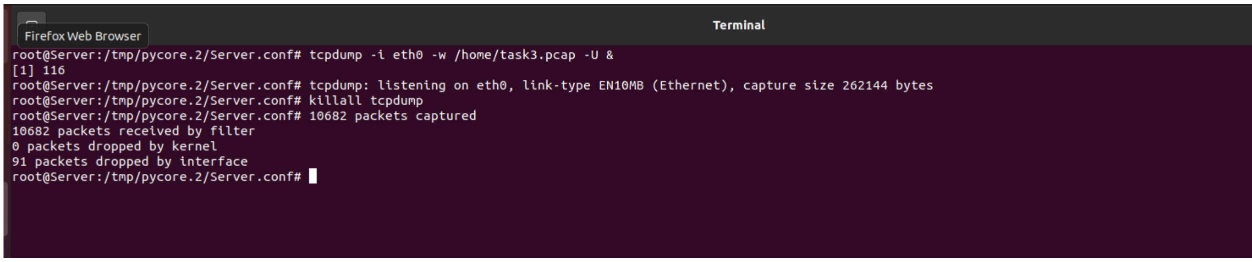


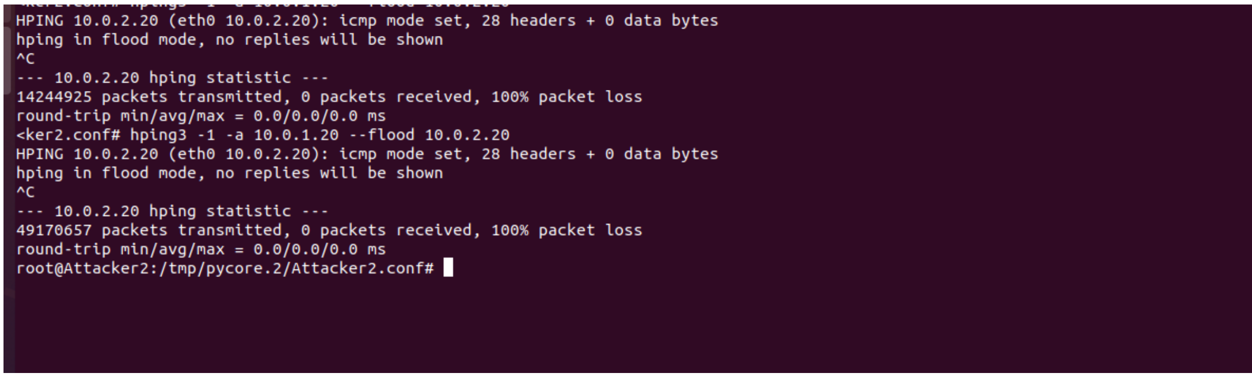
b. What is the maximum attack magnitude?
c. Show a Wireshark I/O graph from the Server matching your answer. **Use bits/s on the y-axis.**
d. Describe your setup and the resulting I/O graph

3. (1pts) Attacker 2 runs a DoS ICMP reflection attack on the Server using Attacker 1 as a reflection
    a. Show the exact command you used


task3.pcap
Firefox Web Browser
Capture Analyze Statistics Telephony Wireless Tools Help
Apply a display filter ... <Ctrl-/>
No. Time Source Destination Protocol Length Info
1 0.000000 10.0.1.1 224.0.0.5 OSPF 78 Hello Packet
2 2.003752 10.0.1.1 224.0.0.5 OSPF 78 Hello Packet
3 4.005342 10.0.1.1 224.0.0.5 OSPF 78 Hello Packet
4 5.988090 fe80::200:ff:feaa:2 ff02::5 OSPF 90 Hello Packet
5 6.007562 10.0.1.1 224.0.0.5 OSPF 78 Hello Packet
6 8.008989 10.0.1.1 224.0.0.5 OSPF 78 Hello Packet
7 10.010797 10.0.1.1 224.0.0.5 OSPF 78 Hello Packet
8 11.898224 fe80::90af:fcff:fe3… ff02::2 ICMPv6 70 Router Solicitation from 92:af:fc:3e:23:73
9 12.012808 10.0.1.1 224.0.0.5 OSPF 78 Hello Packet
10 14.014788 10.0.1.1 224.0.0.5 OSPF 78 Hello Packet
11 14.193727 10.0.2.20 10.0.1.20 ICMP 42 Echo (ping) reply id=0x3900, seq=0/0, ttl=63
12 14.201885 10.0.2.20 10.0.1.20 ICMP 42 Echo (ping) reply id=0x3900, seq=256/1, ttl=63
13 14.218683 10.0.2.20 10.0.1.20 ICMP 42 Echo (ping) reply id=0x3900, seq=512/2, ttl=63
14 14.227072 10.0.2.20 10.0.1.20 ICMP 42 Echo (ping) reply id=0x3900, seq=768/3, ttl=63
15 14.235484 10.0.2.20 10.0.1.20 ICMP 42 Echo (ping) reply id=0x3900, seq=1024/4, ttl=63
16 14.252275 10.0.2.20 10.0.1.20 ICMP 42 Echo (ping) reply id=0x3900, seq=1280/5, ttl=63
17 14.260672 10.0.2.20 10.0.1.20 ICMP 42 Echo (ping) reply id=0x3900, seq=1536/6, ttl=63
18 14.269074 10.0.2.20 10.0.1.20 ICMP 42 Echo (ping) reply id=0x3900, seq=1792/7, ttl=63
19 14.285873 10.0.2.20 10.0.1.20 ICMP 42 Echo (ping) reply id=0x3900, seq=2048/8, ttl=63
20 14.294285 10.0.2.20 10.0.1.20 ICMP 42 Echo (ping) reply id=0x3900, seq=2304/9, ttl=63
21 14.302673 10.0.2.20 10.0.1.20 ICMP 42 Echo (ping) reply id=0x3900, seq=2560/10, ttl=63
22 14.319475 10.0.2.20 10.0.1.20 ICMP 42 Echo (ping) reply id=0x3900, seq=2816/11, ttl=63
23 14.327872 10.0.2.20 10.0.1.20 ICMP 42 Echo (ping) reply id=0x3900, seq=3072/12, ttl=63
24 14.336272 10.0.2.20 10.0.1.20 ICMP 42 Echo (ping) reply id=0x3900, seq=3328/13, ttl=63
25 14.353074 10.0.2.20 10.0.1.20 ICMP 42 Echo (ping) reply id=0x3900, seq=3584/14, ttl=63
26 14.361471 10.0.2.20 10.0.1.20 ICMP 42 Echo (ping) reply id=0x3900, seq=3840/15, ttl=63
27 14.369874 10.0.2.20 10.0.1.20 ICMP 42 Echo (ping) reply id=0x3900, seq=4096/16, ttl=63
28 14.386676 10.0.2.20 10.0.1.20 ICMP 42 Echo (ping) reply id=0x3900, seq=4352/17, ttl=63
29 14.395073 10.0.2.20 10.0.1.20 ICMP 42 Echo (ping) reply id=0x3900, seq=4608/18, ttl=63
30 14.403472 10.0.2.20 10.0.1.20 ICMP 42 Echo (ping) reply id=0x3900, seq=4864/19, ttl=63
31 14.420286 10.0.2.20 10.0.1.20 ICMP 42 Echo (ping) reply id=0x3900, seq=5120/20, ttl=63
32 14.428702 10.0.2.20 10.0.1.20 ICMP 42 Echo (ping) reply id=0x3900, seq=5376/21, ttl=63
33 14.437111 10.0.2.20 10.0.1.20 ICMP 42 Echo (ping) reply id=0x3900, seq=5632/22, ttl=63
34 14.453900 10.0.2.20 10.0.1.20 ICMP 42 Echo (ping) reply id=0x3900, seq=5888/23, ttl=63
35 14.462302 10.0.2.20 10.0.1.20 ICMP 42 Echo (ping) reply id=0x3900, seq=6144/24, ttl=63
36 14.470702 10.0.2.20 10.0.1.20 ICMP 42 Echo (ping) reply id=0x3900, seq=6400/25, ttl=63
Frame 1: 78 bytes on wire (624 bits), 78 bytes captured (624 bits)
Ethernet II, Src: 00:00:00_aa:00:02 (00:00:00:aa:00:02), Dst: IPv4mcast_05 (01:00:5e:00:00:05)
Internet Protocol Version 4, Src: 10.0.1.1, Dst: 224.0.0.5
Open Shortest Path First

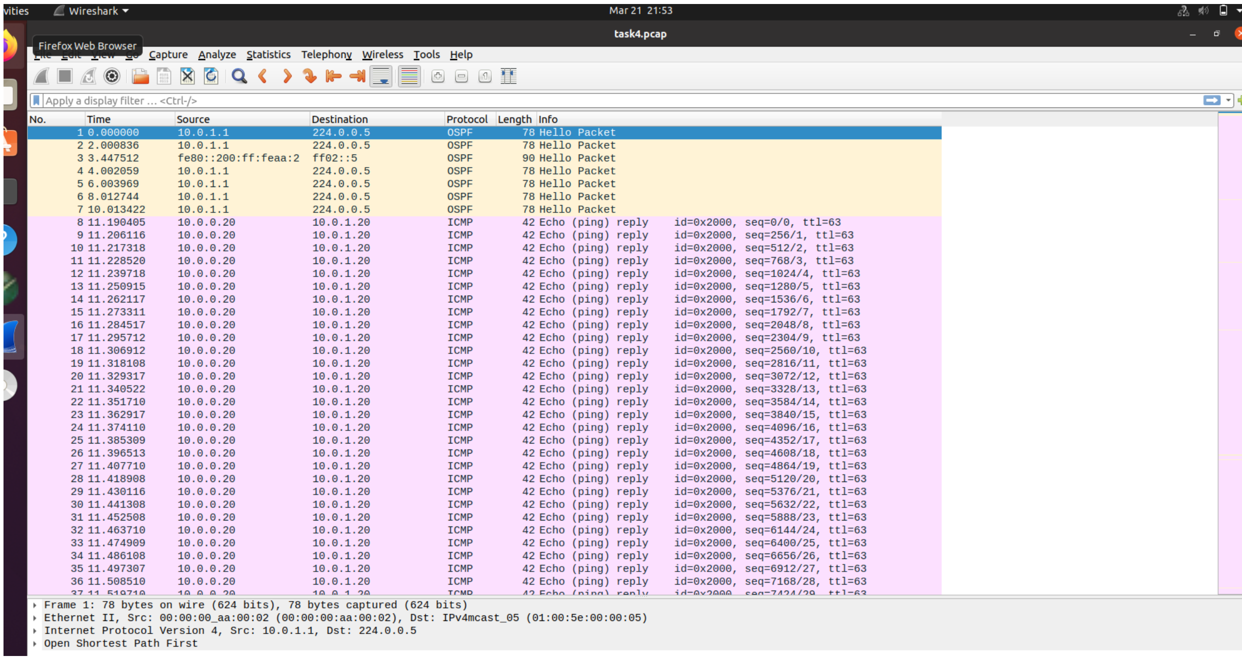

Wireshark · I/O Graphs · task3.pcap
Wireshark I/O Graphs: task3.pcap
60000
50000
40000
30000
20000
10000
0
60
80
100
120
Time (s)
All Packets
Filtered packets
Filtered packets
Click to select packet 3993 (65s = 91).
Enabled Graph Name Display Filter Color Style Y Axis Y Field SMA Period
All Packets Line Packets None
TCP Errors tcp.analysis… Bar Packets None
Filtered pa… Line Bits None
Filtered pa… tcp.flags.sy… Line Packets None
Mouse drags zooms
Interval 1 sec
Time of day
Log scale
Reset
Help
Copy
Copy from
Close
Save As…





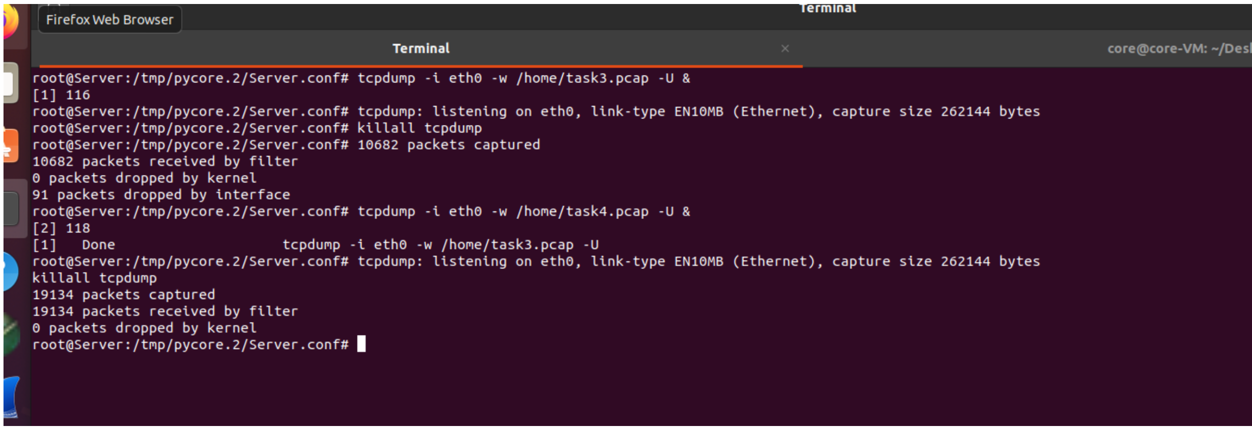
b. What is the maximum attack magnitude?
c. Show a Wireshark I/O graph from the Server matching your answer. **Use bits/s on the y-axis.**
d. Describe your setup and the resulting I/O graph

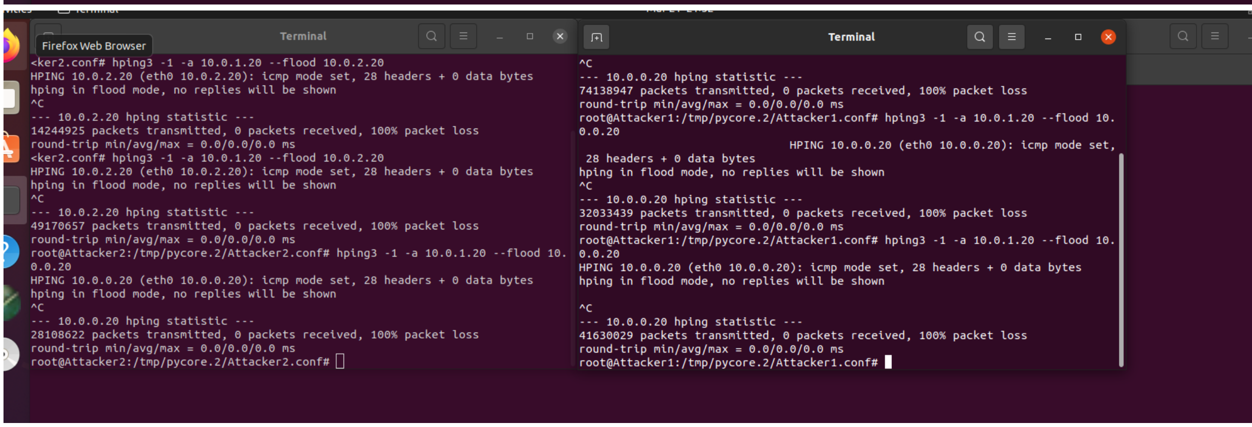
4. (1pts) Attacker1 and Attacker2 both simultaneously run DoS ICMP reflection attacks on the Server using Client as a reflection
   a. Show the exact command you used



Wireshark
Mar 21 21:53
task4.pcap
Firefox Web Browser
Capture Analyze Statistics Telephony Wireless Tools Help
Apply a display filter ... <Ctrl-/>
No. Time Source Destination Protocol Length Info
1 0.000000 10.0.1.1 224.0.0.5 OSPF 78 Hello Packet
2 2.000836 10.0.1.1 224.0.0.5 OSPF 78 Hello Packet
3 3.447512 fe80::200:ff:feaa:2 ff02::5 OSPF 90 Hello Packet
4 4.002059 10.0.1.1 224.0.0.5 OSPF 78 Hello Packet
5 6.003969 10.0.1.1 224.0.0.5 OSPF 78 Hello Packet
6 8.012744 10.0.1.1 224.0.0.5 OSPF 78 Hello Packet
7 10.013422 10.0.1.1 224.0.0.5 OSPF 78 Hello Packet
8 11.190405 10.0.0.20 10.0.1.20 ICMP 42 Echo (ping) reply id=0x2000, seq=0/0, ttl=63
9 11.206116 10.0.0.20 10.0.1.20 ICMP 42 Echo (ping) reply id=0x2000, seq=256/1, ttl=63
10 11.217318 10.0.0.20 10.0.1.20 ICMP 42 Echo (ping) reply id=0x2000, seq=512/2, ttl=63
11 11.228520 10.0.0.20 10.0.1.20 ICMP 42 Echo (ping) reply id=0x2000, seq=768/3, ttl=63
12 11.239718 10.0.0.20 10.0.1.20 ICMP 42 Echo (ping) reply id=0x2000, seq=1024/4, ttl=63
13 11.250915 10.0.0.20 10.0.1.20 ICMP 42 Echo (ping) reply id=0x2000, seq=1280/5, ttl=63
14 11.262117 10.0.0.20 10.0.1.20 ICMP 42 Echo (ping) reply id=0x2000, seq=1536/6, ttl=63
15 11.273311 10.0.0.20 10.0.1.20 ICMP 42 Echo (ping) reply id=0x2000, seq=1792/7, ttl=63
16 11.284517 10.0.0.20 10.0.1.20 ICMP 42 Echo (ping) reply id=0x2000, seq=2048/8, ttl=63
17 11.295712 10.0.0.20 10.0.1.20 ICMP 42 Echo (ping) reply id=0x2000, seq=2304/9, ttl=63
18 11.306912 10.0.0.20 10.0.1.20 ICMP 42 Echo (ping) reply id=0x2000, seq=2560/10, ttl=63
19 11.318108 10.0.0.20 10.0.1.20 ICMP 42 Echo (ping) reply id=0x2000, seq=2816/11, ttl=63
20 11.329317 10.0.0.20 10.0.1.20 ICMP 42 Echo (ping) reply id=0x2000, seq=3072/12, ttl=63
21 11.340522 10.0.0.20 10.0.1.20 ICMP 42 Echo (ping) reply id=0x2000, seq=3328/13, ttl=63
22 11.351710 10.0.0.20 10.0.1.20 ICMP 42 Echo (ping) reply id=0x2000, seq=3584/14, ttl=63
23 11.362917 10.0.0.20 10.0.1.20 ICMP 42 Echo (ping) reply id=0x2000, seq=3840/15, ttl=63
24 11.374110 10.0.0.20 10.0.1.20 ICMP 42 Echo (ping) reply id=0x2000, seq=4096/16, ttl=63
25 11.385309 10.0.0.20 10.0.1.20 ICMP 42 Echo (ping) reply id=0x2000, seq=4352/17, ttl=63
26 11.396513 10.0.0.20 10.0.1.20 ICMP 42 Echo (ping) reply id=0x2000, seq=4608/18, ttl=63
27 11.407710 10.0.0.20 10.0.1.20 ICMP 42 Echo (ping) reply id=0x2000, seq=4864/19, ttl=63
28 11.418908 10.0.0.20 10.0.1.20 ICMP 42 Echo (ping) reply id=0x2000, seq=5120/20, ttl=63
29 11.430116 10.0.0.20 10.0.1.20 ICMP 42 Echo (ping) reply id=0x2000, seq=5376/21, ttl=63
30 11.441308 10.0.0.20 10.0.1.20 ICMP 42 Echo (ping) reply id=0x2000, seq=5632/22, ttl=63
31 11.452508 10.0.0.20 10.0.1.20 ICMP 42 Echo (ping) reply id=0x2000, seq=5888/23, ttl=63
32 11.463710 10.0.0.20 10.0.1.20 ICMP 42 Echo (ping) reply id=0x2000, seq=6144/24, ttl=63
33 11.474909 10.0.0.20 10.0.1.20 ICMP 42 Echo (ping) reply id=0x2000, seq=6400/25, ttl=63
34 11.486108 10.0.0.20 10.0.1.20 ICMP 42 Echo (ping) reply id=0x2000, seq=6656/26, ttl=63
35 11.497307 10.0.0.20 10.0.1.20 ICMP 42 Echo (ping) reply id=0x2000, seq=6912/27, ttl=63
36 11.508510 10.0.0.20 10.0.1.20 ICMP 42 Echo (ping) reply id=0x2000, seq=7168/28, ttl=63
Frame 1: 78 bytes on wire (624 bits), 78 bytes captured (624 bits)
Ethernet II, Src: 00:00:00_aa:00:02 (00:00:00:aa:00:02), Dst: IPv4mcast_05 (01:00:5e:00:00:05)
Internet Protocol Version 4, Src: 10.0.1.1, Dst: 224.0.0.5
Open Shortest Path First

task4.pcap
Firefox Web Browser
Capture Analyze Statistics Telephony Wireless Tools Help
Wireshark · I/O Graphs · task4.pcap
Wireshark I/O Graphs: task4.pcap
150000
100000
50000
0
0
50
100
Time (s)
Click to select packet 10710 (80s = 208).
Enabled Graph Name Display Filter Color Style Y Axis Y Field SMA Period
All Packets Line Packets None
TCP Errors tcp.analysis… Bar Packets None
Filtered pa… Line Bits None
Filtered pa… tcp.flags.sy… Line Packets None
Mouse drags zooms
Interval 1 sec
Help





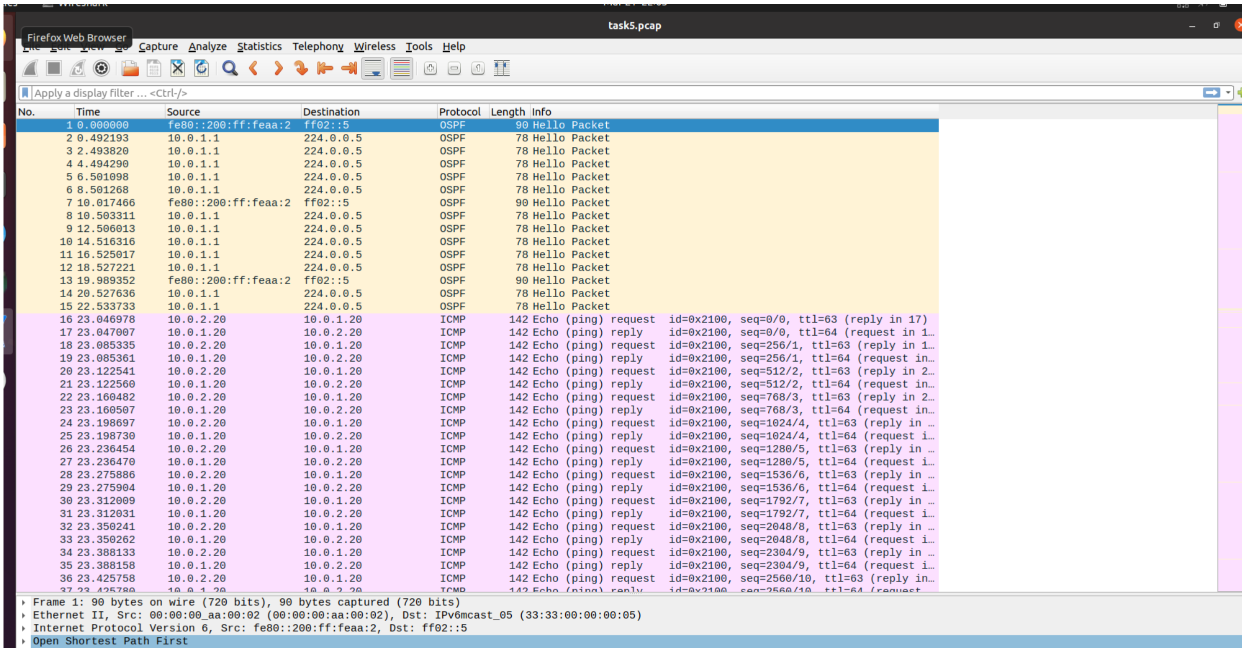
b. What is the maximum attack magnitude?

c. Show a Wireshark I/O graph from the Server matching your answer. **Use bits/s on the y-axis.**

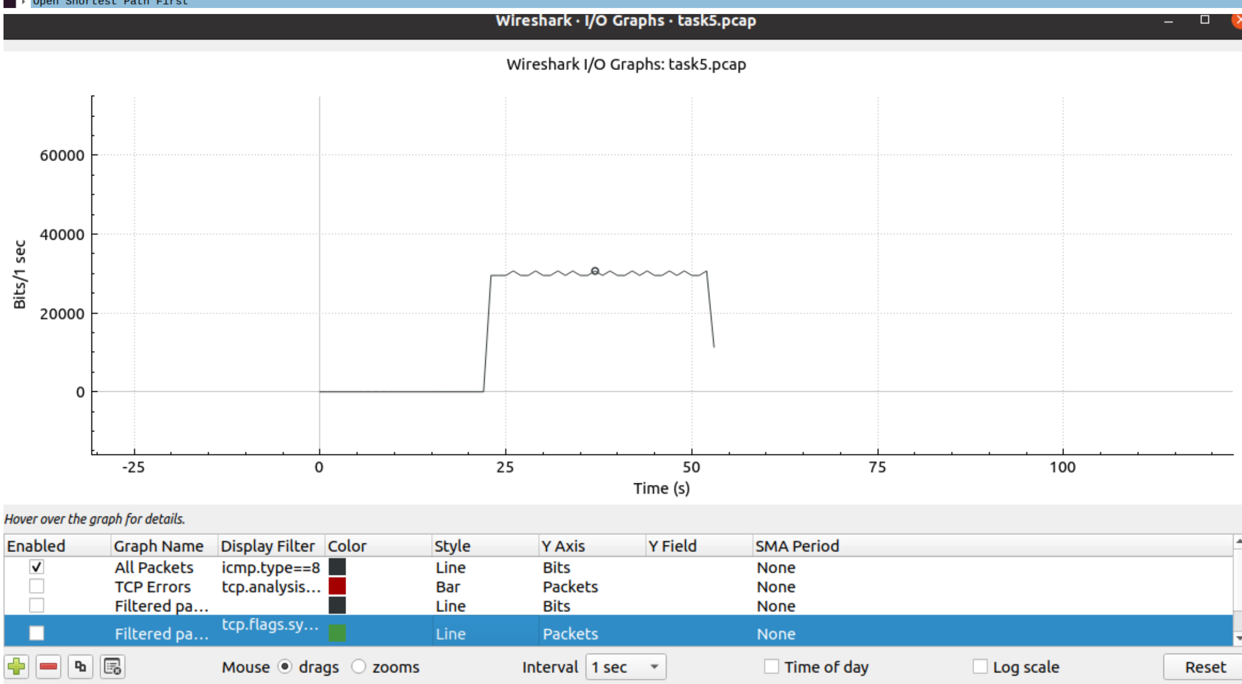
d. Describe your setup and the resulting I/O graph

5. (1pts) Attacker 1 runs a DoS ICMP attack on the Server. Do not use the --flood option. We will play with the -d option (data size) and --interval option (packet rate) to tune the magnitude of the attack

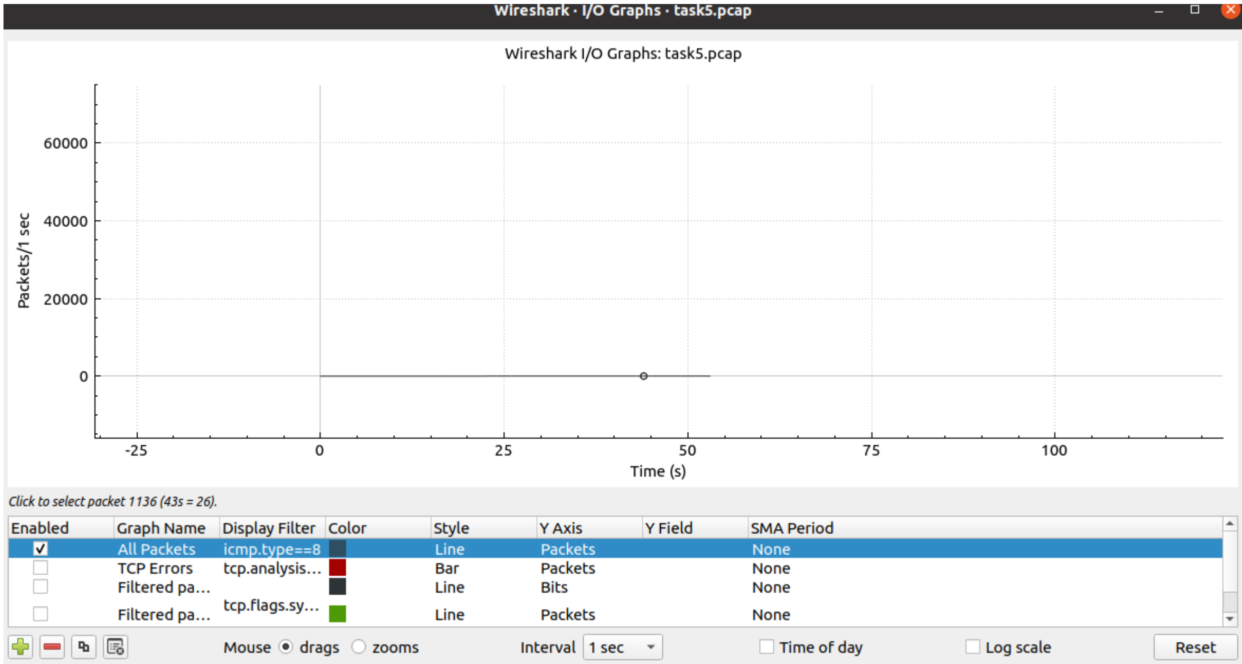
   a. Show the exact command you used



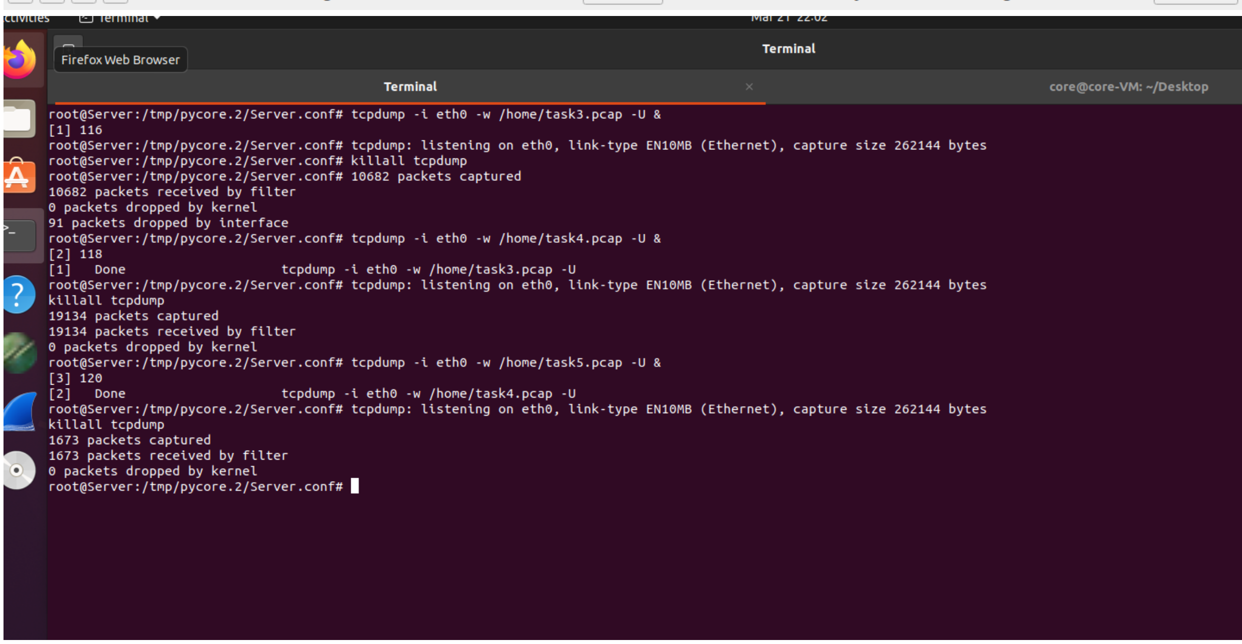
task5.pcap
Firefox Web Browser
Capture Analyze Statistics Telephony Wireless Tools Help
Apply a display filter … <Ctrl-/>
No. Time Source Destination Protocol Length Info
1 0.000000 fe80::200:ff:feaa:2 ff02::5 OSPF 90 Hello Packet
2 0.492193 10.0.1.1 224.0.0.5 OSPF 78 Hello Packet
3 2.493820 10.0.1.1 224.0.0.5 OSPF 78 Hello Packet
4 4.494290 10.0.1.1 224.0.0.5 OSPF 78 Hello Packet
5 6.501098 10.0.1.1 224.0.0.5 OSPF 78 Hello Packet
6 8.501268 10.0.1.1 224.0.0.5 OSPF 78 Hello Packet
7 10.017466 fe80::200:ff:feaa:2 ff02::5 OSPF 90 Hello Packet
8 10.503311 10.0.1.1 224.0.0.5 OSPF 78 Hello Packet
9 12.506013 10.0.1.1 224.0.0.5 OSPF 78 Hello Packet
10 14.516316 10.0.1.1 224.0.0.5 OSPF 78 Hello Packet
11 16.525017 10.0.1.1 224.0.0.5 OSPF 78 Hello Packet
12 18.527221 10.0.1.1 224.0.0.5 OSPF 78 Hello Packet
13 19.989352 fe80::200:ff:feaa:2 ff02::5 OSPF 90 Hello Packet
14 20.527636 10.0.1.1 224.0.0.5 OSPF 78 Hello Packet
15 22.533733 10.0.1.1 224.0.0.5 OSPF 78 Hello Packet
16 23.046978 10.0.2.20 10.0.1.20 ICMP 142 Echo (ping) request id=0x2100, seq=0/0, ttl=63 (reply in 17)
17 23.047007 10.0.1.20 10.0.2.20 ICMP 142 Echo (ping) reply id=0x2100, seq=0/0, ttl=64 (request in 1…
18 23.085335 10.0.2.20 10.0.1.20 ICMP 142 Echo (ping) request id=0x2100, seq=256/1, ttl=63 (reply in 1…
19 23.085361 10.0.1.20 10.0.2.20 ICMP 142 Echo (ping) reply id=0x2100, seq=256/1, ttl=64 (request in…
20 23.122541 10.0.2.20 10.0.1.20 ICMP 142 Echo (ping) request id=0x2100, seq=512/2, ttl=63 (reply in 2…
21 23.122560 10.0.1.20 10.0.2.20 ICMP 142 Echo (ping) reply id=0x2100, seq=512/2, ttl=64 (request in…
22 23.160482 10.0.2.20 10.0.1.20 ICMP 142 Echo (ping) request id=0x2100, seq=768/3, ttl=63 (reply in 2…
23 23.160507 10.0.1.20 10.0.2.20 ICMP 142 Echo (ping) reply id=0x2100, seq=768/3, ttl=64 (request in…
24 23.198697 10.0.2.20 10.0.1.20 ICMP 142 Echo (ping) request id=0x2100, seq=1024/4, ttl=63 (reply in …
25 23.198730 10.0.1.20 10.0.2.20 ICMP 142 Echo (ping) reply id=0x2100, seq=1024/4, ttl=64 (request i…
26 23.236454 10.0.2.20 10.0.1.20 ICMP 142 Echo (ping) request id=0x2100, seq=1280/5, ttl=63 (reply in …
27 23.236470 10.0.1.20 10.0.2.20 ICMP 142 Echo (ping) reply id=0x2100, seq=1280/5, ttl=64 (request i…
28 23.275886 10.0.2.20 10.0.1.20 ICMP 142 Echo (ping) request id=0x2100, seq=1536/6, ttl=63 (reply in …
29 23.275904 10.0.1.20 10.0.2.20 ICMP 142 Echo (ping) reply id=0x2100, seq=1536/6, ttl=64 (request i…
30 23.312009 10.0.2.20 10.0.1.20 ICMP 142 Echo (ping) request id=0x2100, seq=1792/7, ttl=63 (reply in …
31 23.312031 10.0.1.20 10.0.2.20 ICMP 142 Echo (ping) reply id=0x2100, seq=1792/7, ttl=64 (request i…
32 23.350241 10.0.2.20 10.0.1.20 ICMP 142 Echo (ping) request id=0x2100, seq=2048/8, ttl=63 (reply in …
33 23.350262 10.0.1.20 10.0.2.20 ICMP 142 Echo (ping) reply id=0x2100, seq=2048/8, ttl=64 (request i…
34 23.388133 10.0.2.20 10.0.1.20 ICMP 142 Echo (ping) request id=0x2100, seq=2304/9, ttl=63 (reply in …
35 23.388158 10.0.1.20 10.0.2.20 ICMP 142 Echo (ping) reply id=0x2100, seq=2304/9, ttl=64 (request i…
36 23.425758 10.0.2.20 10.0.1.20 ICMP 142 Echo (ping) request id=0x2100, seq=2560/10, ttl=63 (reply in…
Frame 1: 90 bytes on wire (720 bits), 90 bytes captured (720 bits)
Ethernet II, Src: 00:00:00_aa:00:02 (00:00:00:aa:00:02), Dst: IPv6mcast_05 (33:33:00:00:00:05)
Internet Protocol Version 6, Src: fe80::200:ff:feaa:2, Dst: ff02::5
Open Shortest Path First



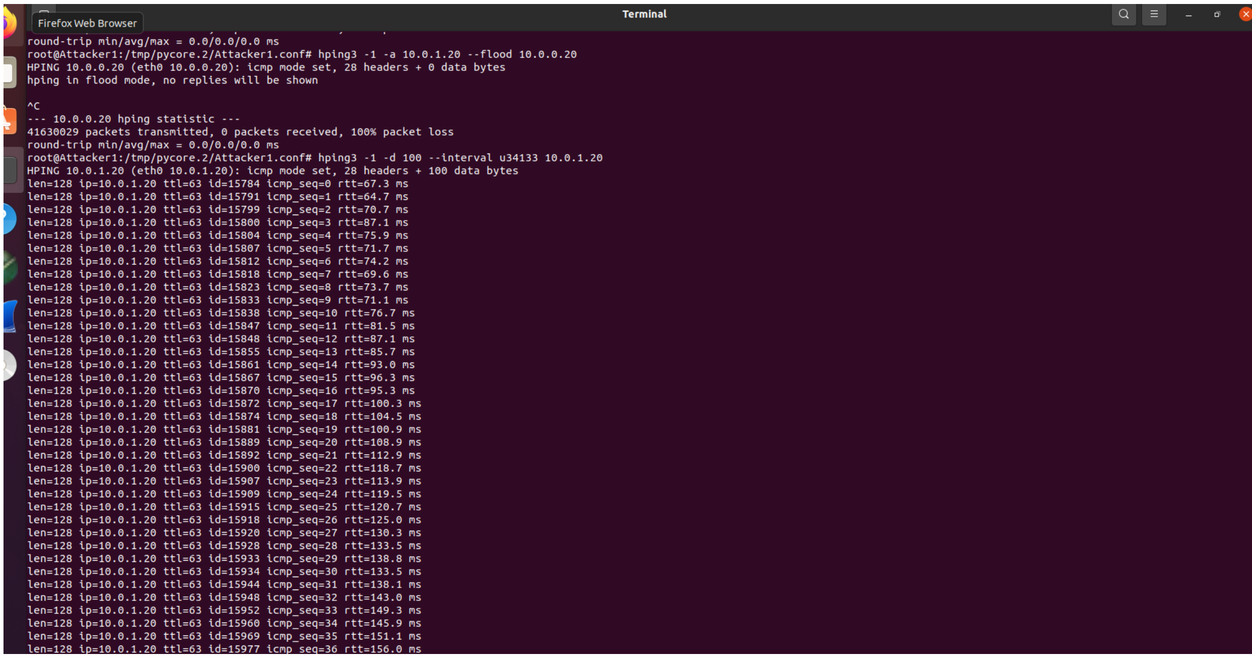
Wireshark · I/O Graphs · task5.pcap
Wireshark I/O Graphs: task5.pcap
60000
40000
20000
0
Bits/1 sec
-25
0
25
50
75
100
Time (s)
Hover over the graph for details.
Enabled Graph Name Display Filter Color Style Y Axis Y Field SMA Period
All Packets icmp.type==8 Line Bits None
TCP Errors tcp.analysis… Bar Packets None
Filtered pa… Line Bits None
Filtered pa… tcp.flags.sy… Line Packets None
Mouse drags zooms
Interval 1 sec
Time of day
Log scale
Reset



Wireshark · I/O Graphs · task5.pcap
Wireshark I/O Graphs: task5.pcap
Packets/1 sec
60000
40000
20000
0
-25
0
25
50
75
100
Time (s)
Click to select packet 1136 (43s = 26).
Enabled Graph Name Display Filter Color Style Y Axis Y Field SMA Period
All Packets icmp.type==8 Line Packets None
TCP Errors tcp.analysis… Bar Packets None
Filtered pa… Line Bits None
Filtered pa… tcp.flags.sy… Line Packets None
Mouse drags zooms
Interval 1 sec
Time of day
Log scale
Reset



Terminal
Firefox Web Browser
Terminal
core@core-VM: ~/Desktop
root@Server:/tmp/pycore.2/Server.conf# tcpdump -i eth0 -w /home/task3.pcap -U &
[1] 116
root@Server:/tmp/pycore.2/Server.conf# tcpdump: listening on eth0, link-type EN10MB (Ethernet), capture size 262144 bytes
root@Server:/tmp/pycore.2/Server.conf# killall tcpdump
root@Server:/tmp/pycore.2/Server.conf# 10682 packets captured
10682 packets received by filter
0 packets dropped by kernel
91 packets dropped by interface
root@Server:/tmp/pycore.2/Server.conf# tcpdump -i eth0 -w /home/task4.pcap -U &
[2] 118
[1] Done tcpdump -i eth0 -w /home/task3.pcap -U
root@Server:/tmp/pycore.2/Server.conf# tcpdump: listening on eth0, link-type EN10MB (Ethernet), capture size 262144 bytes
killall tcpdump
19134 packets captured
19134 packets received by filter
0 packets dropped by kernel
root@Server:/tmp/pycore.2/Server.conf# tcpdump -i eth0 -w /home/task5.pcap -U &
[3] 120
[2] Done tcpdump -i eth0 -w /home/task4.pcap -U
root@Server:/tmp/pycore.2/Server.conf# tcpdump: listening on eth0, link-type EN10MB (Ethernet), capture size 262144 bytes
killall tcpdump
1673 packets captured
1673 packets received by filter
0 packets dropped by kernel
root@Server:/tmp/pycore.2/Server.conf#



b. Using a -d of 100 bytes, what --interval is needed to get a to 30Kbps attack magnitude?

c. Prove your answer using two I/O graphs: **one using packets/s on the y-axis and the other using bits/s.**

d. Describe your setup and the resulting I/O graph